

$f: M \rightarrow M \rightarrow \text{surjection}$
 $\uparrow$ (injection)
 Noetherian (Artinian) $\rightarrow f$ is iso

$$M/\ker f \cong M$$

$$[Ler T^n \cap Im T^n]$$

$$|L| \leq T^n$$

$$0 \xrightarrow{f} L \xrightarrow{g} M \xrightarrow{g} N \rightarrow 0$$

$$M : \mathcal{H}(E) \Leftrightarrow L, N : \mathcal{H}(E)$$

$$(Art', n) \quad (Art', n)$$

$$M/f(L) \cong N$$

$$\mathbb{M}_L \cong N$$

Artinian ring

$$d \subset \subset$$

$$\mathbb{Z} \subset \mathbb{Z}^2 \subset \mathbb{Z}^3 \subset \dots$$

$$0 \rightarrow L \xrightarrow{f} M \hookrightarrow f^* \mathcal{I}_Y$$

$$M \xrightarrow{g} N \rightarrow 0 \hookrightarrow g^* \mathcal{I}_X$$

(Cor) $M_i \in \mathcal{C}_1 \quad (1 \leq i \leq n)$
 $\Rightarrow \bigoplus_{i=1}^n M_i \in \mathcal{C}_1$

$$0 \rightarrow M_1 \rightarrow \bigoplus_{i=1}^n M_i \rightarrow \bigoplus_{i=1}^{n-1} M_i \rightarrow 0$$

$T: V \rightarrow V$ is diagonalizable $\Leftrightarrow m_T$ splits and separable

" triangulizable $\Leftrightarrow m_T$ is separable

E/F is separable $\Leftrightarrow \# \text{Emb}_F(E, \bar{F}) = [E:F]$

E/F is splitting $\Leftrightarrow \text{Emb}_F(E, \bar{F}) = \text{Aut}(E/F)$

relation $\{\}$